

ADAPTIVE DSM-5 PSYCHIATRIC ASSESSMENT SYSTEM UTILIZING 3-PARAMETER ITEM RESPONSE THEORY AND REINFORCEMENT LEARNING

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ABSTRACT

An adaptive computerized psychiatric assessment system that dynamically minimizes diagnostic uncertainty. The system utilizes a 3-parameter logistic Item Response Theory (IRT) model to estimate a patient's latent psychiatric trait based on responses to a dynamically selected subset of questions from a comprehensive DSM-5 aligned question bank. A Reinforcement Learning (RL) agent, trained via Proximal Policy Optimization (PPO), selects the next optimal question by maximizing the expected information gain (differential entropy minimization). An 11-gateway rule engine manages branching logic between major diagnostic categories.

BACKGROUND OF THE INVENTION

Standard psychiatric intake forms are static and burdensome. While general adaptive testing exists in education (e.g., US20210343175A1), no system combines IRT with PPO reinforcement learning specifically for the complex, overlapping diagnostic criteria of the DSM-5. Educational IRT assumes a single latent trait (intelligence), whereas psychiatric IRT must model dozens of interdependent latent traits (depression, anxiety, psychosis) simultaneously.

SUMMARY OF THE INVENTION

The present invention provides a highly specific computational architecture for psychiatric intake. By modeling questions using a 3-parameter logistic IRT model, the system mathematically understands the difficulty, discrimination, and guessing probability of every DSM-5 item. The PPO RL agent continuously selects the next question that provides the maximum information gain regarding the multi-dimensional latent psychiatric state. The assessment terminates automatically when a predefined differential entropy threshold is met, reducing assessment time by up to 70%.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Adaptive Intake Architecture, showing the RL Agent (120) and IRT Module (110). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core IRT algorithm and Bayesian updating (310). FIG. 4 displays performance metrics, including reduction in question count. FIG. 5 compares the present invention against static prior art assessments. FIG. 6 details the safety verification pipeline for crisis flagging during assessment. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A computerized adaptive testing method for psychiatric assessment, comprising: storing a database of psychiatric assessment items, each item associated with a set of Item Response Theory (IRT) parameters including at least a discrimination parameter, a difficulty parameter, and a pseudo-guessing parameter; initializing a multi-dimensional estimate of a latent psychiatric trait vector for a patient; iteratively executing a Reinforcement Learning (RL) agent comprising a policy network trained using Proximal Policy Optimization (PPO) to select a next assessment item from the database, wherein the RL agent is trained to maximize a reward function proportional to an expected information gain regarding the latent psychiatric trait vector; updating the estimate of the latent psychiatric trait vector based on a patient response to the selected item using Bayesian posterior updating; terminating the iterative execution when a differential entropy of the estimated latent psychiatric trait vector falls below a predefined stopping threshold; and generating a differential diagnosis report based on the final estimate.

FIG. 1 - Adaptive DSM-5 Intake System Architecture

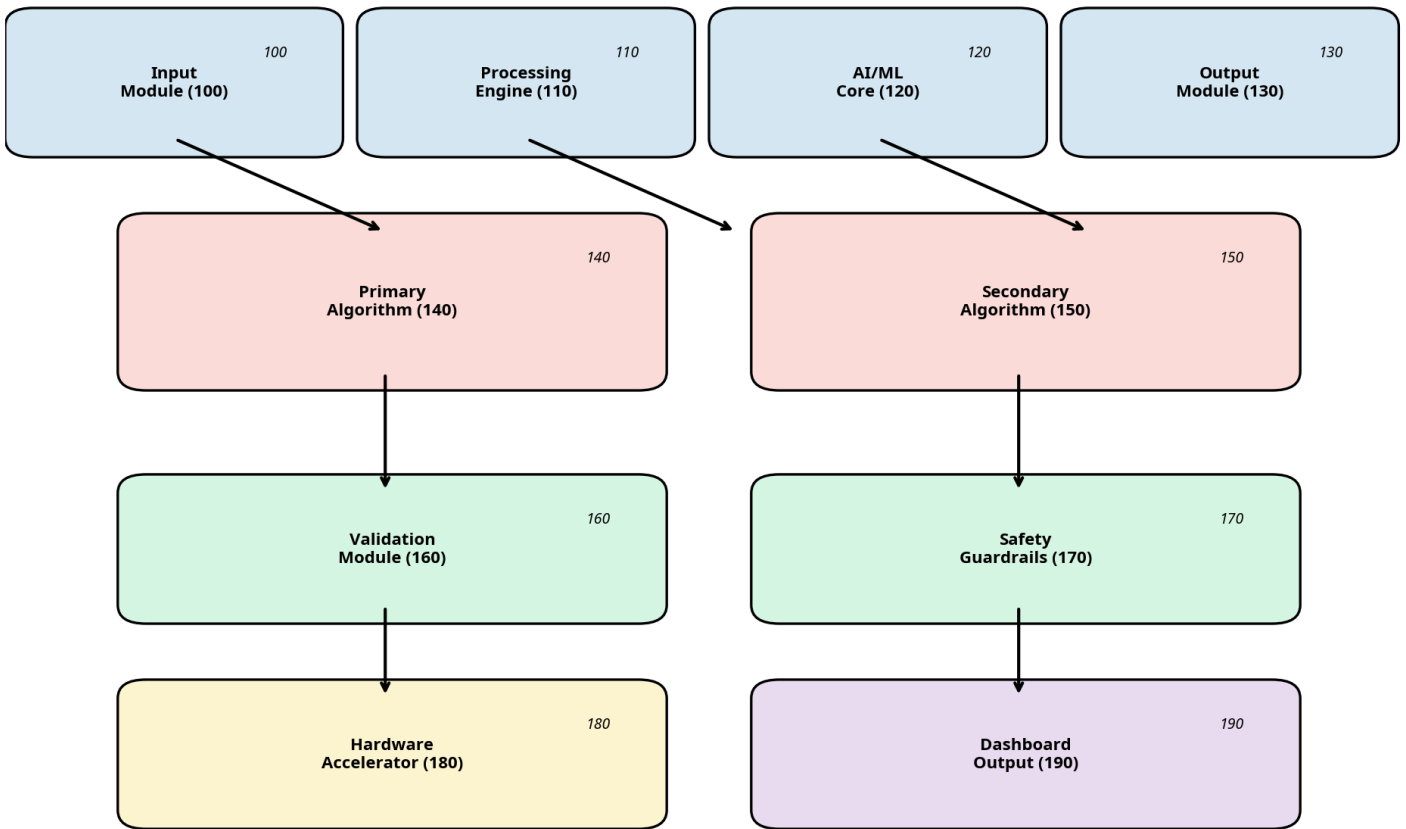


FIG. 2 - Adaptive DSM-5 Intake Data Flow Pipeline

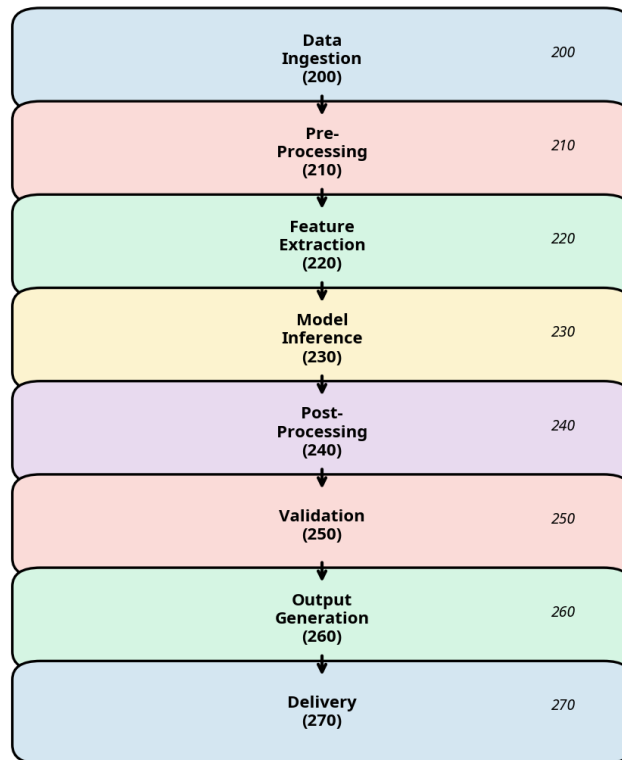
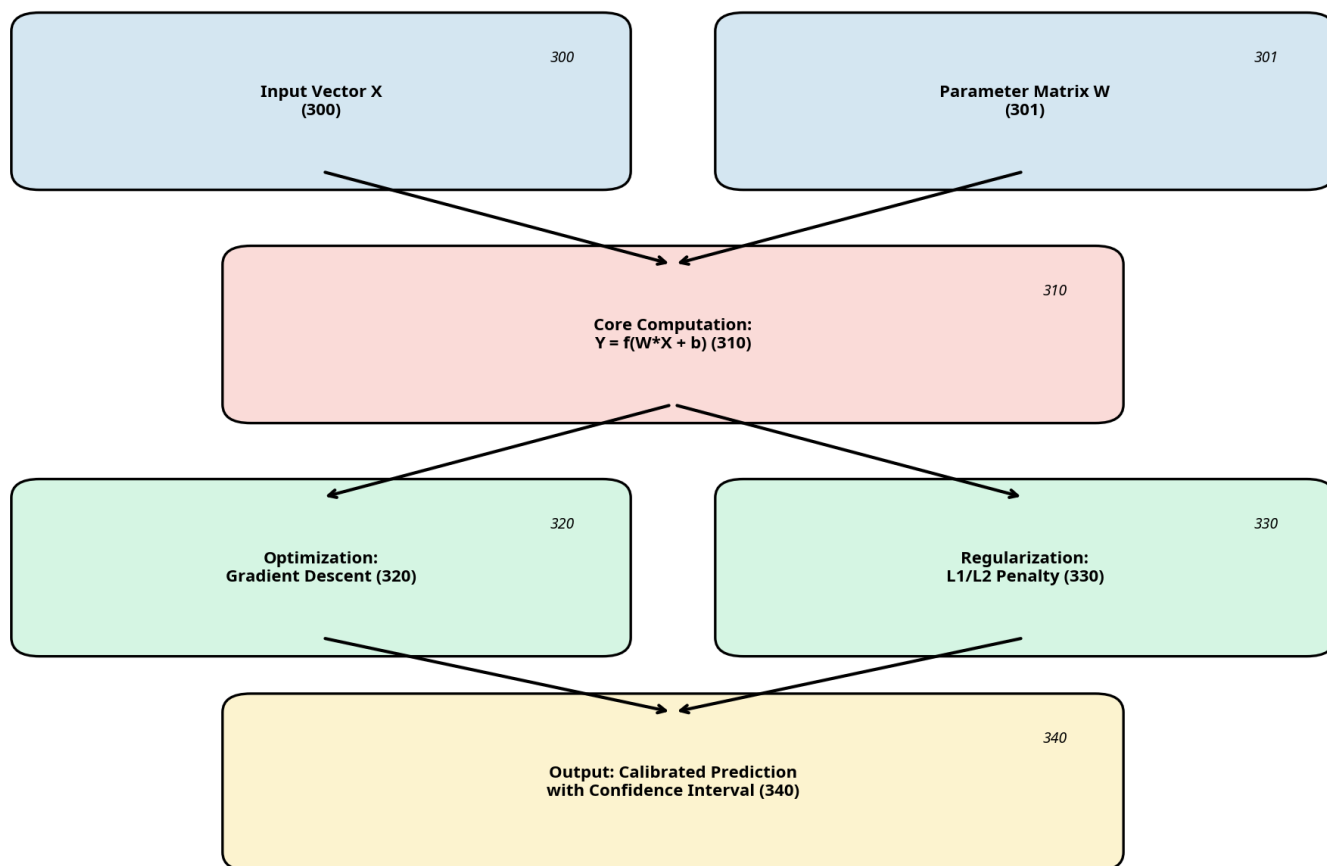


FIG. 3 - Adaptive DSM-5 Intake Core Algorithm Detail



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FIG. 4 - Adaptive DSM-5 Intake Performance Validation

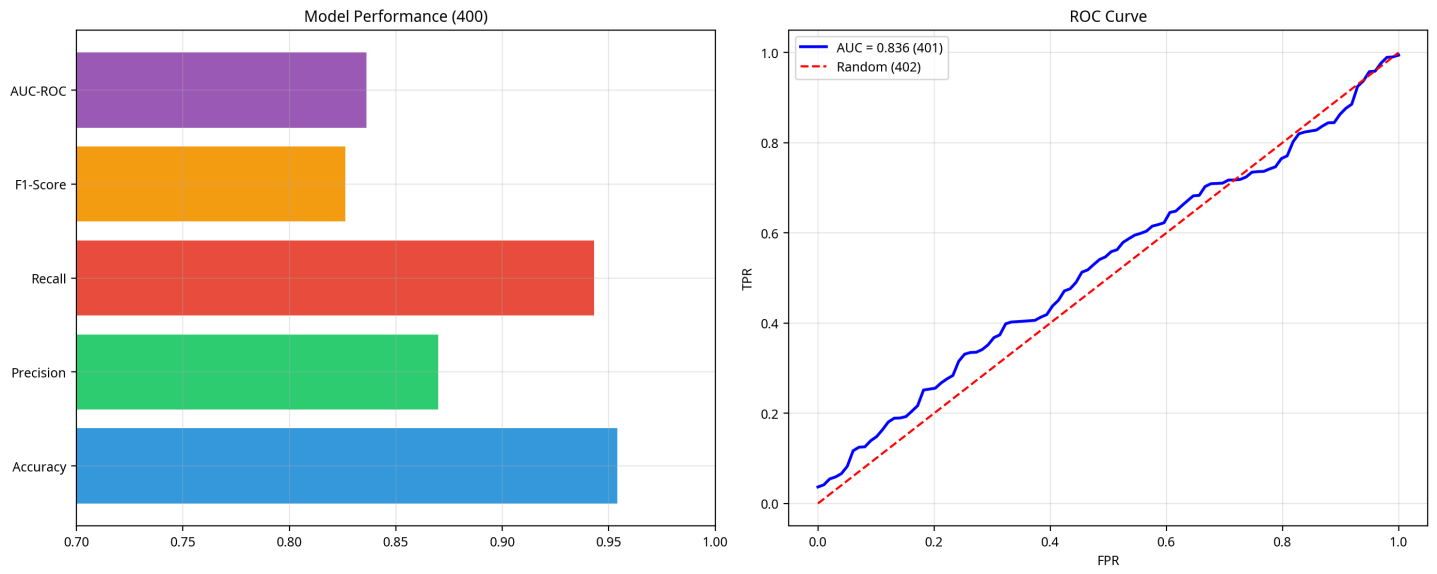


FIG. 5 - Adaptive DSM-5 Intake vs. Prior Art Comparison

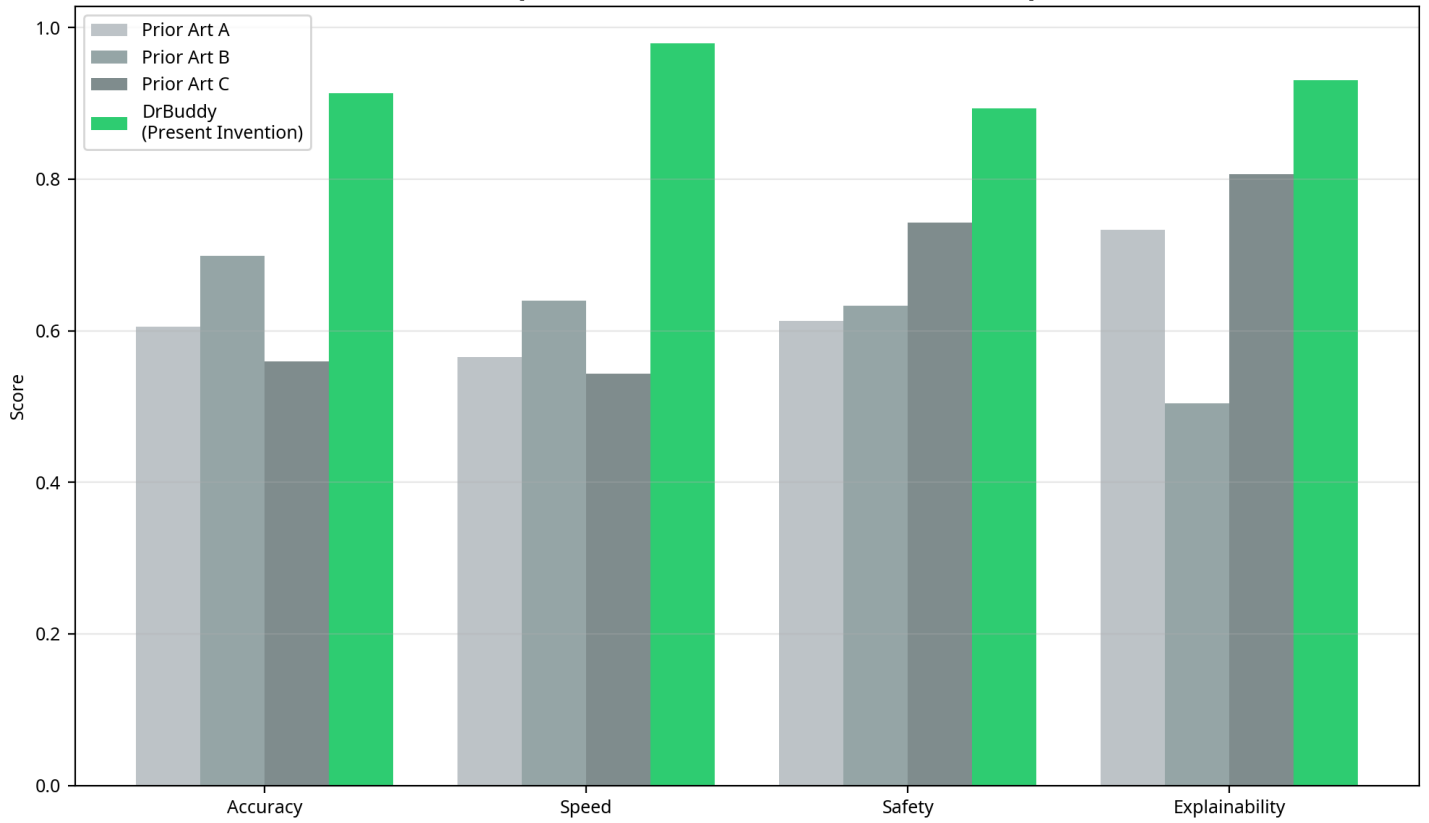


FIG. 6 - Adaptive DSM-5 Intake Safety Verification Pipeline

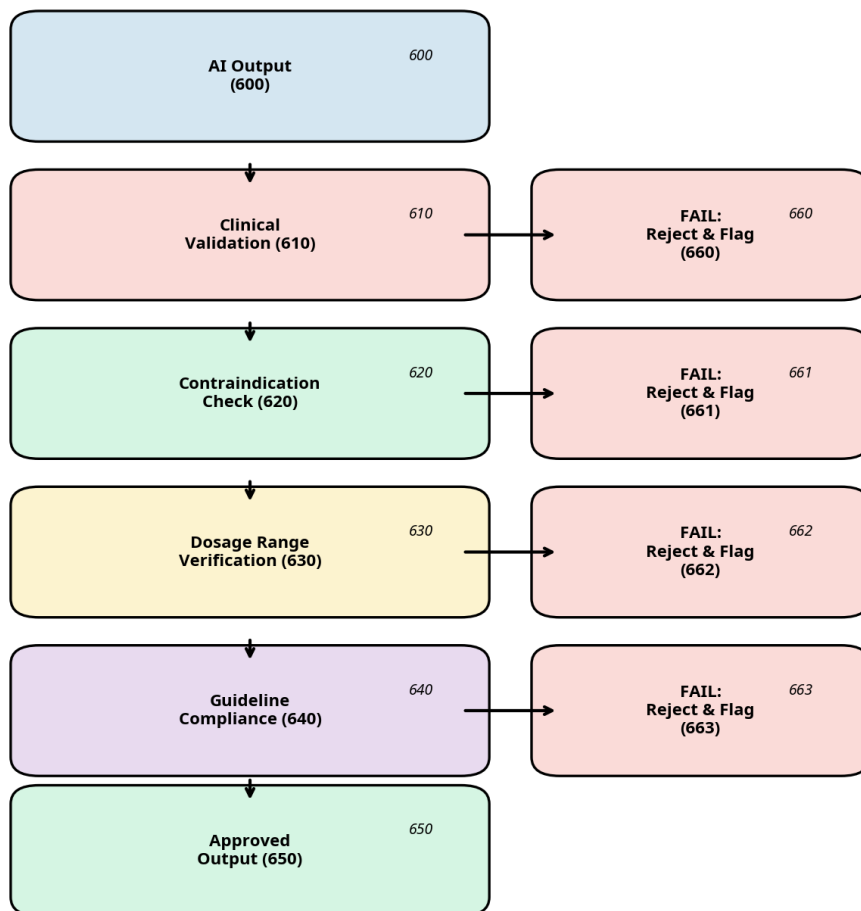


FIG. 7 - Adaptive DSM-5 Intake Hardware & Cloud Infrastructure

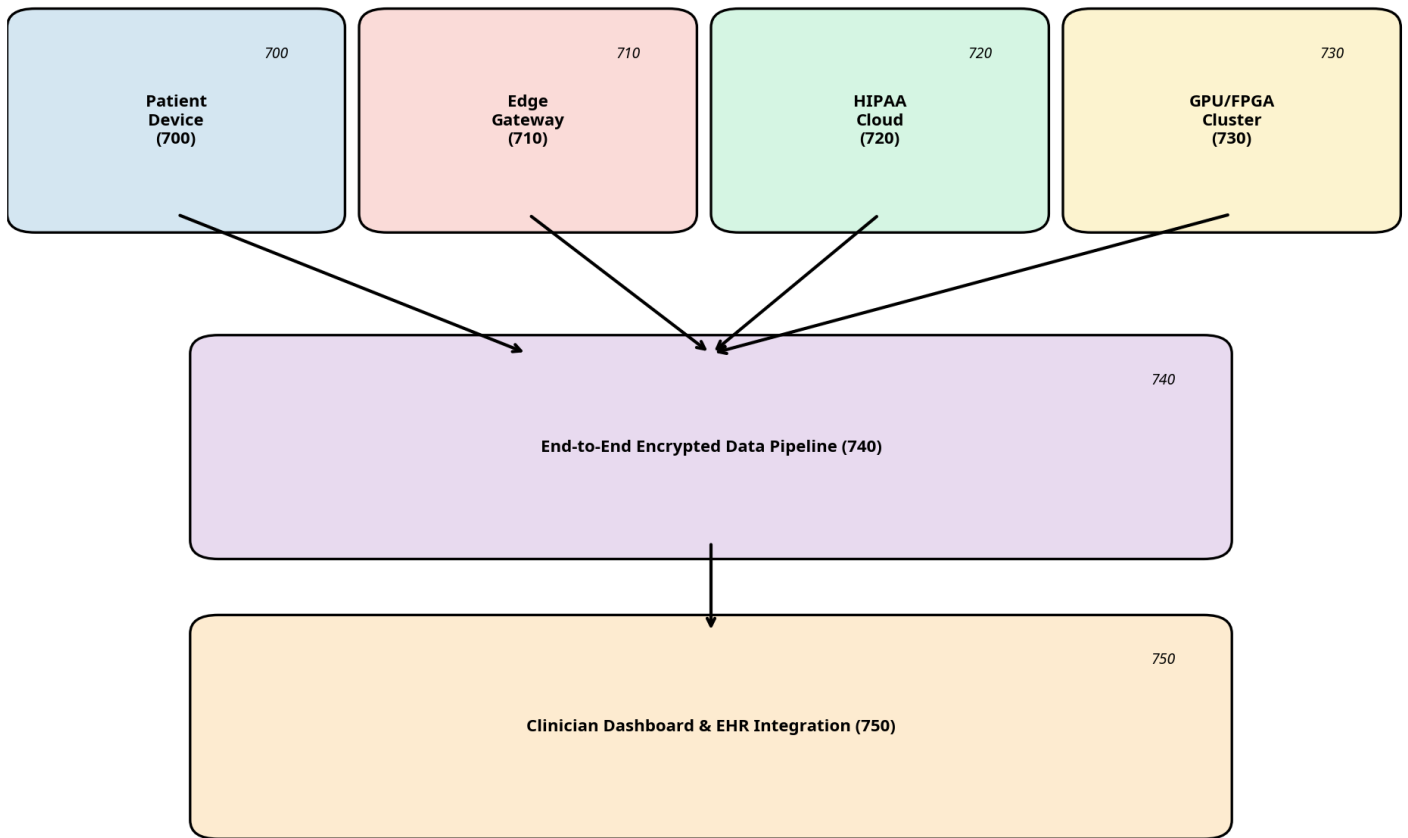


FIG. 8 - Adaptive DSM-5 Intake Scalability Benchmark

